

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light greenish-blue. They are positioned diagonally, with the blue one in front of the green one.

Utilizing software to assign and account for tasks

Jakub Turkowski 280288

Why do we need task allocation and accounting?

Task allocation and accounting helps with:

- project planning
- efficiency and order within the project
- visualisation of responsibilities
- tracking project progress
- teamwork



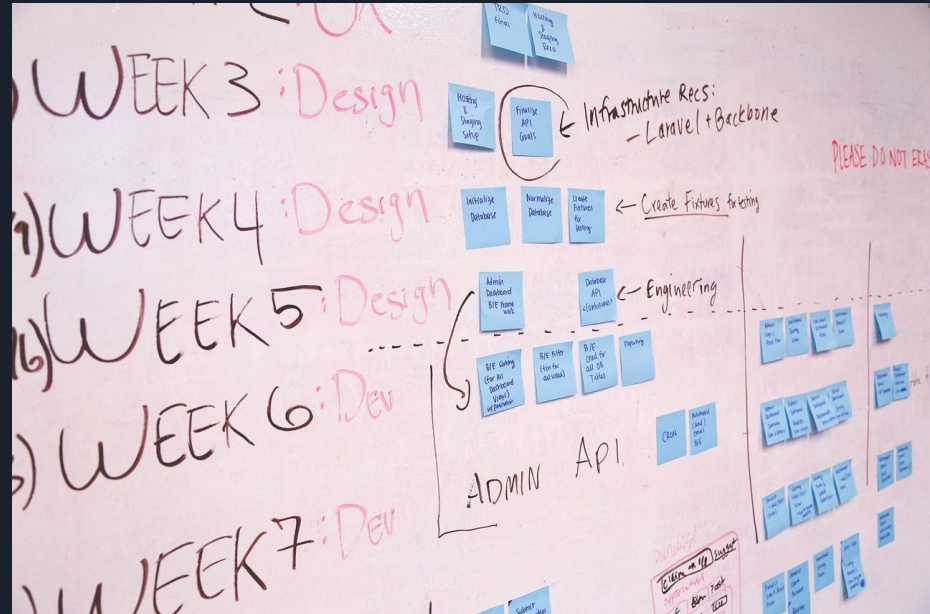
A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Specification of functionalities

User needs

As a project manager I should be able to:

- create and organize tasks
- assign tasks to team members
- plan and monitor task execution
- set task requirements and due date
- track the progress of tasks
- track time spent on tasks



Software functionalities

No.	Functionality name	Input	Output	Purpose	Significance
1	Task creation	Task name	Created task	To have tasks to allocate and account for	Must have
2	Add task description	Description of responsibilities	Task with detailed requirements	To make it clear what is required for task completion	Must have
3	Categorize tasks	Estimated time, difficulty, required resources	Task with assigned parameters	To support better task allocation decisions (e.g. in terms of team members)	Should have
4	Assign users to a task	Selected users	Task with assigned users	To assign tasks to specific team members	Must have



Software functionalities, cont.

5	Set task deadlines	A deadline date	Task with an expected completion date	To specify when the task is supposed to be completed by	Must have
6	Update task completion status	Completion status (e.g. in progress, completed)	Task updated with the completion status	To keep track of the progress made across tasks	Must have
7	Log time spent on specific tasks	Time spent on a task	A task with elapsed completion time	To measure effort and productivity	Should have



Software functionalities, cont.

8	Set task priority	Priority level	Prioritized task	To better communicate which tasks should be done first	Must have
9	Notifications	Task deadline	Notification	To remind people working on tasks of impending deadlines	Could have
10	Generate reports on task progress and time usage	Task data and time logs	A report	To analyze the outcome of specific tasks and improving the allocation and accounting processes	Should have



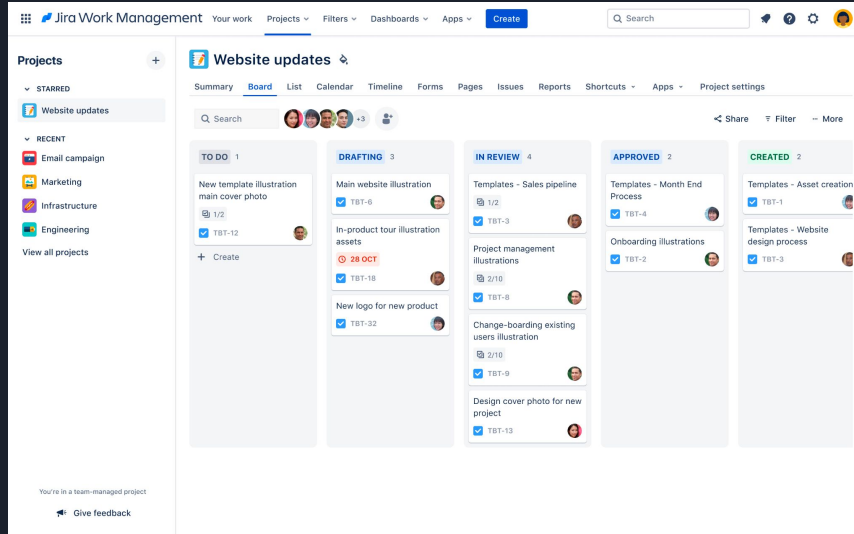
Software functionalities, cont.

11	Show update history	Changes to the tasks	History of updates (to task or entire project)	To better visualize the progress over time	Should have
12	Add task dependencies	Dependency (task 1 depends on task 2)	A task dependent on another task completion	To define relationships between tasks and ensure correct execution order	Must have
13	View the task timeline	Task deadlines, start times and dependencies	Visual timeline (e.g. Gantt chart)	To better see how far into the project we are and notice potential problems	Should have

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Comparative price analysis

Jira



Producer: Atlassian

Type: Web

Free tier: available

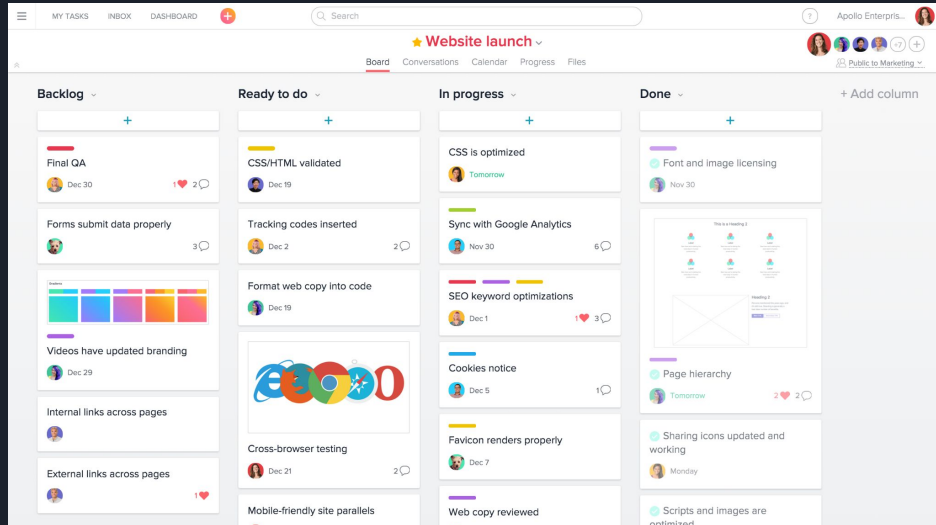
Premium price: \$7.91 for standard, \$14.54 for premium, enterprise tier

The paid features are mostly collaboration-related, also things like more storage, users, or uptime

Main strength: very robust

Main weakness: difficult to learn

Asana



Producer: Asana Inc.

Type: Web

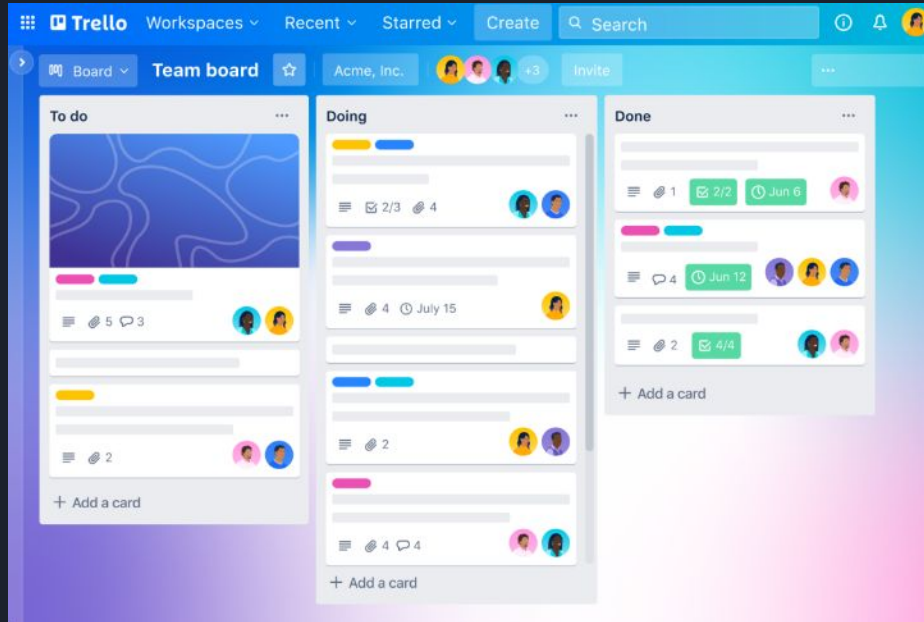
Free tier: available

Premium price: 10.99 € for starter,
24.99 € for premium, enterprise tier
Unlimited reporting, more users,
some features that are already in the
free version but with limitations

Main strength: strong collaboration
features

Main weakness: limited capabilities of
free version

Trello



Producer: Atlassian

Type: Web

Free tier: available

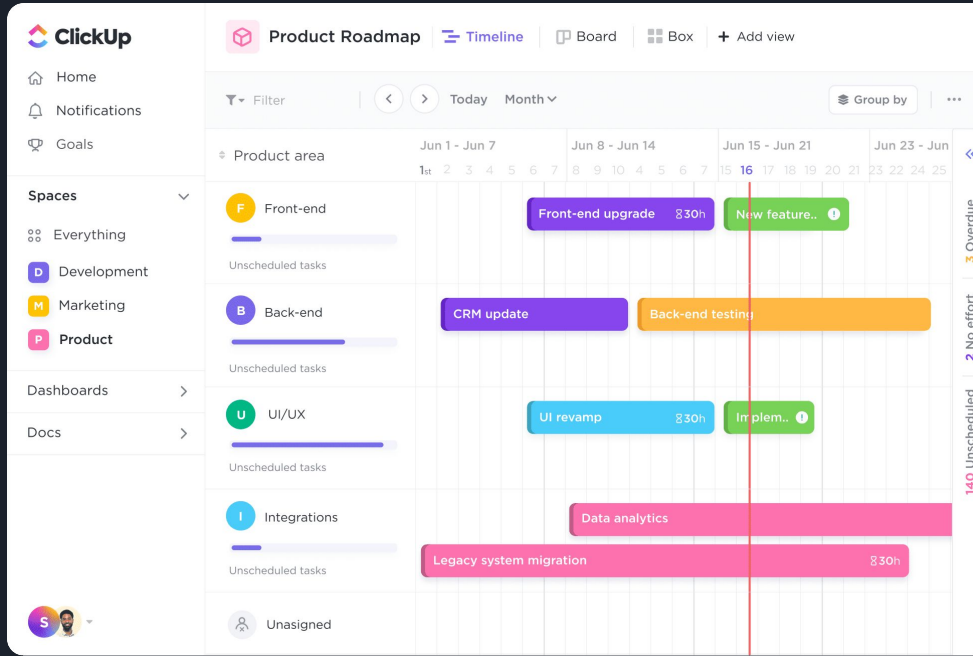
Premium price: \$5 for standard, \$10 for premium, enterprise solution \$17.38 - 7.38 (depending on amount of users)

Custom fields, AI tools, storage, advanced checklists, collapsing lists

Main strength: extremely simple

Main weakness: not suitable for big projects

ClickUp



Producer: ClickUp

Type: Web

Free tier: available

Premium price: \$7.91 for standard, \$12 for business, enterprise solution

Unlimited Gantt charts and custom fields, unlimited reporting, more storage, integration with google apps

Main strength: many different features

Main weakness: limitations on the free plan



	Create tasks	Add task description	Categorize tasks	Assign users to a task	Set task deadlines
Jira	Tasks can be created in the backlog	Descriptions can be added to tasks	Tasks can be assigned story points, tags, custom fields can be created	Users can be assigned to tasks	Deadlines can be added, tasks can also be added to specific sprints
Asana	Tasks can be created	Tasks can have descriptions and subtasks	Tags can be added to tasks, subtasks can be added, custom fields can be created	Users can be assigned to tasks	Tasks can have due dates
Trello	Created in form of cards	Cards can have descriptions	Tags can be assigned to tasks	Users can be assigned to cards	Cards can have due dates
ClickUp	Tasks can be created	Tasks can have descriptions and subtasks	Tags can be added to tasks, subtasks can be added, custom fields can be added	Users can be assigned to tasks	Tasks can have due dates

	Update task completion status	Log time spent on specific tasks	Set task priority
Jira	Can be updated directly for the task or by moving it to a list to a preexisting or custom completion level	Can be deduced from update history, upgradable with plugins (?)	Priority can be chosen for tasks
Asana	Can be updated for the task as complete or not complete, additional statuses can be created by sections or custom fields	Can be deduced from update history, available in paid version	Priority can be chosen for tasks, custom fields can be added
Trello	Cards can be marked as complete / not complete, more detailed statuses can be created with lists	Can be deduced from update history, upgradable with plugins (?)	With tags in free version, custom fields in paid
ClickUp	Can be updated directly for the task or by moving it to a list to a preexisting or custom completion level	Dedicated timer for each task	Priority can be chosen for tasks

	Generate reports on task progress and time usage	Show update history	Add task dependencies	View the task timeline
Jira	Report creation with custom dashboards and charts, separate summary page, more options in paid	For full project and specific tasks	Dependencies can be added directly on the timeline	Timeline available in form of a timeline and calendar
Asana	Report creation with custom dashboards and charts, separate summary page, more options in paid	For specific tasks, AI generated summary for the whole project	Dependencies can be added directly on the timeline	Timeline available in form of Gantt chart and calendar, limited in free version
Trello	Not supported	For full project and specific tasks	Not supported	Not supported
ClickUp	Report creation with custom dashboards and charts, separate summary page, more options in paid	For full project and specific tasks	Dependencies can be added directly on the timeline	Timeline available in form of Gantt chart and calendar, limited in free version



Software chosen - ClickUp

- includes a free tier
- it is web-based
- people are familiar with it already
- has every required functionality with only some limitations in the free plan
- no plugins required
- AI agent can help you find features

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with subtle diagonal lines.

Software demonstration



Exemplary task

You're a project manager and you're not sure how to allocate the tasks for an ongoing project. Your project has 3 team members - one senior and 2 juniors. Add a way to categorise the tasks based on task difficulty to make the allocation process easier.



Task for you

For the last report you created a project schedule for a small online store. For this week's task, you will improve your solution.

1. Assign a priority level and a description (could be also in form of subtasks) to each of the created tasks.
2. Think of 2 categories, that you will assign to your tasks - 1 should include metrics helpful in task allocation process (e.g. task difficulty) and the other should be helpful with accounting for tasks (e.g. lines of code written). Implement them in form of a custom field using a dropdown menu or a rating. Assign the task allocation-related category to your tasks.
3. Look at the schedule created for the previous lab. Is there anything you would change considering the new metrics? Justify your new approach (or lack thereof).